

GP-CPS: A Gaussian Process Framework for Data-Driven Digital Twins of Cyber-Physical Systems

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Abstract—Digital twins—virtual replicas of physical systems—enable simulation, prediction, and anomaly detection without interrupting real-world operations. In this paper, we present GP-CPS, a framework that constructs digital twins of Cyber-Physical Systems (CPS) using an ensemble of Gaussian Process (GP) regressors. Our approach learns the one-step transition dynamics from multivariate sensor data, provides principled uncertainty quantification through the GP posterior variance, and detects anomalies when observed behaviour deviates beyond 2σ from the predicted distribution. We evaluate GP-CPS on a Smart Building Zone case study with four interdependent variables (temperature, humidity, CO₂, occupancy) and three anomaly scenarios. Critically, we perform a comprehensive comparison with three standard alternatives—Linear Regression (LR), Random Forest (RF), and Multi-Layer Perceptron (MLP)—on identical data. While all methods achieve similar predictive accuracy ($R^2 > 0.95$), GP-CPS is the *only* method that provides built-in, calibrated uncertainty quantification, enabling principled anomaly detection without a separate model. The framework achieves F1 scores above 0.97 across three anomaly scenarios, trains in under 4 minutes on a single CPU core, and provides interpretable kernel hyperparameters with direct physical meaning. Our results demonstrate that GP-based digital twins offer a unique combination of predictive accuracy, uncertainty quantification, and anomaly detection that alternative methods cannot match.

Index Terms—Cyber-Physical Systems, Digital Twin, Gaussian Process, Anomaly Detection, Uncertainty Quantification, Kernel Methods, Smart Buildings

I. INTRODUCTION

Modern Cyber-Physical Systems (CPS)—tight integrations of computation, communication, and physical processes—are foundational to modern industry, transport, and infrastructure [9], [15]. The proliferation of sensors in the Industrial Internet of Things (IIoT) has enabled massive streams of multivariate time series data that capture the complex, interdependent behaviour of these systems. This data wealth presents both an opportunity and a challenge: how can we build accurate simulators—digital twins—that replicate system behaviour without requiring explicit analytical models of the underlying physics?

Data-driven approaches have emerged as a compelling answer [14]. By learning patterns directly from sensor data, machine learning models can capture nonlinear dynamics, stochastic behaviours, and interactions between variables that

would be prohibitively difficult to model analytically. Generative models, including Variational Autoencoders [7], Generative Adversarial Networks [4], and Diffusion Models [5], have been successfully applied to generate realistic multivariate time series for CPS simulation. However, these approaches require significant computational resources (GPU training) and do not naturally provide uncertainty estimates for their predictions.

Standard regression methods—Linear Regression (LR) [10], Random Forests (RF) [2], and Multi-Layer Perceptrons (MLP) [6]—offer computationally cheaper alternatives that can be trained on CPU in seconds or minutes. These methods have been extensively used in system identification and time series forecasting. However, they share a critical limitation: they provide only point estimates of the system state without quantifying prediction uncertainty. This limitation is fundamental for digital twin applications. Anomaly detection, for instance, requires knowing when observed behaviour deviates beyond what normal variability can explain—a distinction that is impossible without uncertainty quantification.

Gaussian Processes (GPs) [12] address this limitation through their Bayesian non-parametric framework. As a distribution over functions, a GP provides not only a predicted mean but also a predictive variance that quantifies confidence in the prediction. This uncertainty naturally captures both the inherent stochasticity of the system and the epistemic uncertainty arising from limited data. GPs have been successfully applied in system identification [8], control [3], and time series modelling, but their application to digital twin construction for CPS remains underexplored.

In this paper, we introduce **GP-CPS**, a framework that leverages Gaussian Processes to build data-driven digital twins of Cyber-Physical Systems. Our contributions are:

- 1) **A GP-based modelling framework** for learning CPS dynamics from multivariate time series data, using an ensemble of independent GPs—one per output variable—with shared input features that include both current state and control inputs.
- 2) **Uncertainty-based anomaly detection**, where the GP's predictive variance serves as a natural confidence envelope: deviations beyond a calibrated threshold signal

system anomalies without requiring a separate anomaly detection model.

- 3) **A comprehensive comparison** with three standard approaches (Linear Regression, Random Forest, MLP) on identical data, demonstrating that GP-CPS matches their accuracy while providing unique uncertainty quantification capabilities.
- 4) **A detailed case study on a Smart Building Zone**, a multi-variable CPS with nonlinear dynamics coupling temperature, humidity, CO₂, and occupancy, including three distinct anomaly scenarios with quantitative detection metrics.

The remainder of this paper is organised as follows. Section II formulates the problem. Section III provides the necessary background on Gaussian Process regression. Section IV presents the GP-CPS framework in detail. Section V describes the Smart Building Zone case study. Section VI reports experimental results including the comparison with alternative methods. Section VII discusses implications and limitations. Section VIII concludes the paper.

II. PROBLEM FORMULATION

We consider a Cyber-Physical System observed through d sensor variables over discrete time steps $t = 1, \dots, T$. At each time step, the system is characterised by a state vector $\mathbf{s}_t \in \mathbb{R}^d$ and a control input $\mathbf{u}_t \in \mathbb{R}^c$. The system evolves according to an unknown transition function f :

$$\mathbf{s}_{t+1} = f(\mathbf{s}_t, \mathbf{u}_t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\epsilon) \quad (1)$$

where $\boldsymbol{\epsilon}_t$ represents zero-mean Gaussian process noise with covariance $\boldsymbol{\Sigma}_\epsilon$, capturing both intrinsic stochasticity and unmodelled dynamics. This Markovian assumption—that the next state depends only on the current state and control—is standard in system identification [10] and holds approximately when the sampling interval is sufficiently large relative to the system's dominant time constants.

Given a dataset $\mathcal{D} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^{N-1}$ of $N-1$ observed transitions, where:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{s}_t \\ \mathbf{u}_t \end{bmatrix} \in \mathbb{R}^{d+c}, \quad \mathbf{y}_t = \mathbf{s}_{t+1} \in \mathbb{R}^d, \quad (2)$$

our objective is to learn a predictive model $\hat{f}$ such that:

$$\hat{f}(\mathbf{x}) \approx \mathbb{E}[\mathbf{y} | \mathbf{x}], \quad \hat{\sigma}(\mathbf{x}) \approx \sqrt{\mathbb{V}[\mathbf{y} | \mathbf{x}]}, \quad (3)$$

where $\hat{\sigma}$ quantifies the predictive uncertainty. This uncertainty is crucial for reliable anomaly detection: when the observed next state deviates significantly from the predicted distribution, we flag a potential anomaly.

Standard regression methods (LR, RF, MLP) can learn $\hat{f}$ from data, providing point estimates of the expected next state. However, they do not provide $\hat{\sigma}$ —the crucial uncertainty estimate that distinguishes confident predictions from extrapolations. To perform anomaly detection with these methods, one would need to train a separate model (e.g., an autoencoder

or one-class SVM) and tune its detection threshold, adding complexity, computational cost, and an additional source of hyperparameters.

III. GAUSSIAN PROCESS REGRESSION

A Gaussian Process is a collection of random variables, any finite subset of which follows a joint Gaussian distribution [12]. A GP is fully specified by a mean function $m(\mathbf{x})$ and a positive definite covariance function (kernel) $k(\mathbf{x}, \mathbf{x}')$:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')). \quad (4)$$

For a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with input matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^\top$ and target vector $\mathbf{y} = [y_1, \dots, y_N]^\top$, the GP assumes that the observed targets are generated by a latent function f with additive Gaussian noise: $y_i = f(\mathbf{x}_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$.

The joint distribution of the training targets $\mathbf{y}$ and the latent function value at a test point $\mathbf{x}^*$ is:

$$\begin{bmatrix} \mathbf{y} \\ f(\mathbf{x}^*) \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K} + \sigma_n^2 \mathbf{I} & \mathbf{k}_* \\ \mathbf{k}_*^\top & k(\mathbf{x}^*, \mathbf{x}^*) \end{bmatrix}\right), \quad (5)$$

where $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ and $\mathbf{k}_* = [k(\mathbf{x}_1, \mathbf{x}^*), \dots, k(\mathbf{x}_N, \mathbf{x}^*)]^\top$. Conditioning on the observed data yields the predictive distribution:

$$\mu(\mathbf{x}^*) = \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, \quad (6)$$

$$\sigma^2(\mathbf{x}^*) = k(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_*, \quad (7)$$

The predictive mean $\mu(\mathbf{x}^*)$ is a linear combination of the training targets weighted by kernel similarities, while the predictive variance $\sigma^2(\mathbf{x}^*)$ decomposes into prior variance minus the information provided by the training data.

A. Kernel Functions for Physical Systems

The kernel function encodes our assumptions about the function being learned. We consider three widely-used kernels appropriate for modelling physical system dynamics.

a) *Matérn Kernel* ($\nu = 3/2$): This kernel assumes functions that are once-differentiable, which is appropriate for physical systems where velocities exist but accelerations may not:

$$k_M(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{3}r}{\ell}\right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right) + \sigma_n^2 \delta_{\mathbf{x}, \mathbf{x}'}, \quad (8)$$

where $r = \|\mathbf{x} - \mathbf{x}'\|$, ℓ is the characteristic length scale, σ^2 the signal variance, and σ_n^2 the noise variance. The length scale ℓ controls how far one must move in input space for the function value to change significantly—a short ℓ indicates sensitive dynamics, while a long ℓ indicates slow-varying behaviour.

b) *Squared Exponential (RBF) Kernel*:: The RBF kernel assumes infinitely smooth functions:

$$k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{r^2}{2\ell^2}\right) + \sigma_n^2 \delta_{\mathbf{x}, \mathbf{x}'}. \quad (9)$$

c) *Rational Quadratic Kernel*:: This kernel models functions at multiple length scales through the parameter α :

$$k_{\text{RQ}}(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{r^2}{2\alpha\ell^2} \right)^{-\alpha} + \sigma_n^2 \delta_{\mathbf{x}, \mathbf{x}'}. \quad (10)$$

B. Hyperparameter Learning

The hyperparameters $\theta = \{\ell, \sigma^2, \sigma_n^2\}$ are learned by maximising the log marginal likelihood:

$$\log p(\mathbf{y} | \mathbf{X}, \theta) = -\frac{1}{2} \mathbf{y}^\top \mathbf{K}_y^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K}_y| - \frac{N}{2} \log 2\pi, \quad (11)$$

where $\mathbf{K}_y = \mathbf{K} + \sigma_n^2 \mathbf{I}$. This objective automatically balances data-fit (first term) with model complexity (second term), providing a principled regularisation mechanism.

IV. THE GP-CPS FRAMEWORK

A. Architecture

GP-CPS employs an ensemble of d independent Gaussian Process regressors, one for each output dimension $i \in \{1, \dots, d\}$. Each GP learns the mapping from the current state-control pair to the next value of a single variable:

$$y^{(i)} \sim \mathcal{GP}(m_i(\mathbf{x}), k_i(\mathbf{x}, \mathbf{x}')), \quad (12)$$

where m_i is the mean function (set to zero after standardising the data) and k_i is the kernel function for dimension i . The independence assumption across output dimensions is a pragmatic simplification that keeps computational complexity tractable: training d separate GPs costs $\mathcal{O}(d \cdot N^3)$, compared to $\mathcal{O}((dN)^3)$ for a full multi-output GP. For the sizes considered here ($d = 3$, $N \approx 1000$), this is manageable on a single CPU core.

B. Uncertainty-Based Anomaly Detection

Given the GP predictive distribution for each dimension:

$$p(y^{(i)*} | \mathcal{D}, \mathbf{x}^*) = \mathcal{N}(\mu_i^*, \sigma_i^{*2}), \quad (13)$$

we define a multi-variable anomaly score that aggregates normalised deviations across all variables:

$$z(\mathbf{y}^*) = \frac{1}{d} \sum_{i=1}^d \frac{|y^{(i)*} - \mu_i^*|}{\sigma_i^* + \varepsilon}, \quad (14)$$

where ε is a small constant (10^{-10}) to prevent division by zero. An observation is flagged as anomalous when $z(\mathbf{y}^*) > \tau$, where τ is a threshold. Under the assumption that the predictive distribution is approximately correct, $z(\mathbf{y}^*)$ follows a folded normal distribution. Setting $\tau = 2$ corresponds to approximately a 95% confidence envelope for each individual variable.

The key advantage of this approach is its simplicity and theoretical grounding: it does not require a separate anomaly detection model, the GP's uncertainty naturally adapts to regions of the input space with sparse data (providing wider confidence intervals where the model is less certain), and the threshold τ has a clear statistical interpretation.

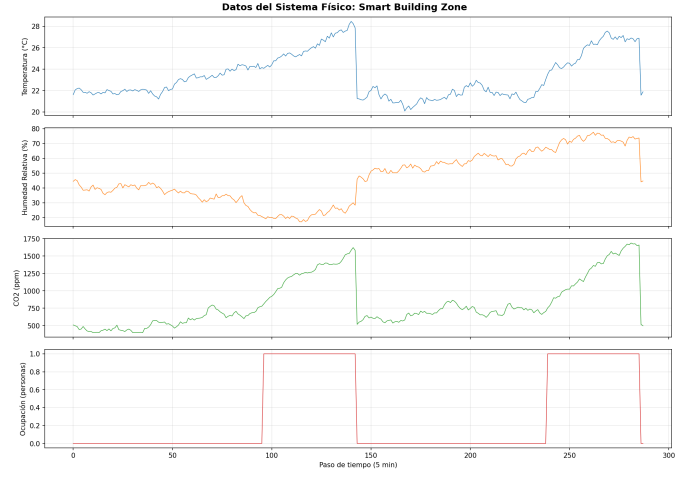


Fig. 1: Multivariate time series from the Smart Building Zone physical model. The four panels show temperature ($^{\circ}\text{C}$), humidity (%), CO_2 concentration (ppm), and occupancy over a 12-hour simulation at 5-minute resolution. The daily schedule is clearly visible: occupancy rises around 8:00, drops at lunch (12:00–13:00), and decreases after 18:00.

C. Multi-Step Simulation

For forward simulation over H steps, we propagate the state recursively. At step h , given the current state $\mathbf{s}_h$ and control $\mathbf{u}_h$, we compute:

$$\hat{\mathbf{s}}_{h+1} = \boldsymbol{\mu}(\mathbf{x}_h), \quad \hat{\boldsymbol{\sigma}}_{h+1} = \boldsymbol{\sigma}(\mathbf{x}_h), \quad (15)$$

where $\boldsymbol{\mu}(\mathbf{x}_h) = [\mu_1(\mathbf{x}_h), \dots, \mu_d(\mathbf{x}_h)]^\top$ and $\boldsymbol{\sigma}(\mathbf{x}_h) = [\sigma_1(\mathbf{x}_h), \dots, \sigma_d(\mathbf{x}_h)]^\top$. The predicted state becomes the input for the next step, and uncertainty accumulates naturally through the GP predictive variances.

D. Algorithm

The complete training and inference procedure is summarised in Algorithm 1.

V. CASE STUDY: SMART BUILDING ZONE

A. System Description

We model an office building zone as a Cyber-Physical System with four sensor variables and two control inputs. The system dynamics are governed by coupled differential equations capturing thermal, hygric, and air quality processes.

1) State Variables:

- T : Temperature ($^{\circ}\text{C}$), physical range $[15, 35]$.
- H : Relative humidity (%), physical range $[10, 90]$.
- C : CO_2 concentration (ppm), physical range $[400, 5000]$.
- O : Occupancy (number of people), discrete $\{0, 1, \dots, 5\}$.

2) Control Inputs:

- P_h : Heater power, normalised to $[0, 1]$.
- V : Ventilation rate, normalised to $[0, 1]$.

Algorithm 1 GP-CPS: Training and Inference

Require: Training data $\mathcal{D} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^N$
Require: Kernel type $k \in \{\text{Matern}, \text{RBF}, \text{RQ}\}$
Require: Anomaly threshold τ (default: 2.0)
Ensure: Predictive model with uncertainty quantification

Training phase

Standardise input features $\mathbf{X} \leftarrow \text{StandardScaler}(\mathbf{X})$
for $i = 1$ to d **do**
 Initialise GP_i with kernel k and train data $\{(\mathbf{x}_t, y_t^{(i)})\}$
 Optimise $\theta_i \leftarrow \arg \max \log p(\mathbf{y}^{(i)} | \mathbf{X}, \theta)$
end for

Prediction phase (at test point $\mathbf{x}^*$)

for $i = 1$ to d **do**
 $(\mu_i^*, \sigma_i^{*2}) \leftarrow \text{GP}_i(\mathbf{x}^*)$
end for
return $\boldsymbol{\mu}^* = [\mu_1^*, \dots, \mu_d^*]^\top$, $\boldsymbol{\sigma}^* = [\sigma_1^*, \dots, \sigma_d^*]^\top$

Anomaly detection (at observed state $\mathbf{y}^*$)

$z \leftarrow \frac{1}{d} \sum_{i=1}^d |y^{(i)*} - \mu_i^*| / (\sigma_i^* + \varepsilon)$
if $z > \tau$ **then**
 Flag observation as **anomalous**
else
 Flag observation as **nominal**
end if

B. Dynamical Model

The system evolves according to coupled differential equations. The temperature dynamics capture heating, heat loss to the exterior, body heat from occupants, and ventilation heat exchange:

$$\frac{dT}{dt} = \frac{1}{C_T} \left[\alpha P_h - \beta(T - T_{\text{ext}}) - \gamma O(T - T_{\text{body}}) - \delta V(T - 18) \right] + \epsilon_T, \quad (16)$$

Humidity is affected by moisture production from occupants, ventilation exchange, and material absorption:

$$\frac{dH}{dt} = \frac{1}{C_H} \left[\eta O(H_{\text{body}} - H) - \theta V(H - H_{\text{ext}}) - \lambda(H - 50) \right] + \epsilon_H, \quad (17)$$

CO₂ concentration is driven by occupant respiration and ventilation renewal:

$$\frac{dC}{dt} = \frac{1}{C_C} \left[\mu O C_{\text{body}} - \nu V(C - C_{\text{ext}}) \right] + \epsilon_C, \quad (18)$$

where $\epsilon_T \sim \mathcal{N}(0, 0.05)$, $\epsilon_H \sim \mathcal{N}(0, 0.3)$, and $\epsilon_C \sim \mathcal{N}(0, 5)$ are process noise terms.

Table I lists the physical parameter values used in simulation.

TABLE I: Physical parameters of the Smart Building Zone dynamical model.

Parameter	Symbol	Value
Thermal capacity	C_T	100.0
Humidity capacity	C_H	50.0
CO ₂ capacity	C_C	80.0
Heater efficiency	α	0.8
Heat loss coefficient	β	0.05
Body heat transfer	γ	0.02
Ventilation heat loss	δ	0.03
Moisture per occupant	η	0.04
Ventilation humidity exchange	θ	0.06
Material absorption rate	λ	0.01
CO ₂ production per occupant	μ	0.005
Ventilation CO ₂ renewal rate	ν	0.05
Exterior temperature	T_{ext}	15.0
Body temperature	T_{body}	33.0
Exterior humidity	H_{ext}	60.0
Exterior CO ₂	C_{ext}	420.0

C. Dataset Generation

Training data is generated by simulating 10 independent scenarios of 12 hours each (144 time steps at 5-minute resolution), with initial parameters perturbed by Gaussian noise ($\sigma = 10\%$ of nominal value) to ensure diversity. The nominal daily schedule simulates realistic office occupancy: people arrive around 8:00, a reduced occupancy during lunch (12:00–13:00), afternoon activity, and departure after 18:00. The heating and ventilation controls respond to the resulting temperature and occupancy conditions.

Validation uses 2 held-out scenarios with different random seeds. Additionally, we generate three anomaly scenarios:

- 1) **Heater malfunction:** The heater is stuck at high power ($P_h = 0.9$) regardless of control signal, simulating a stuck valve or relay failure that causes overheating.
- 2) **Ventilation failure:** The ventilation rate drops to near-zero ($V = 0.02$), simulating a fan or damper failure that causes CO₂ buildup and humidity stagnation.
- 3) **Occupancy spike:** A sudden influx of 5 additional occupants, simulating an unexpected event such as a meeting or gathering that causes rapid changes in all three physical variables.

Figure 1 shows a sample of the generated training data, clearly illustrating the daily cycle and interdependencies between variables.

VI. EXPERIMENTAL RESULTS**A. Experimental Setup**

All experiments were conducted on a single CPU core (AMD) with 16 GB RAM. GP models were implemented using scikit-learn's GaussianProcessRegressor. Input features were standardised to zero mean and unit variance. The Matérn kernel ($\nu = 3/2$) was used as default, and hyperparameters were optimised by maximising the log-marginal-likelihood with 5 random restarts to avoid poor local optima.

For comparison, we trained three alternative models:

TABLE II: Predictive performance comparison across four methods on held-out validation data (286 transitions). Best values in bold.

Metric	LR	RF	MLP	GP-CPS
R^2 (Temperature)	0.955	0.943	0.944	0.957
R^2 (Humidity)	0.976	0.965	0.966	0.977
R^2 (CO ₂)	0.991	0.985	0.986	0.990
Mean R^2	0.974	0.964	0.965	0.975
MAE (Temperature)	0.24	0.29	0.28	0.23
MAE (Humidity)	1.28	1.45	1.42	1.22
MAE (CO ₂ , ppm)	19.4	24.8	23.9	20.7
Training time (s)	0.02	1.40	0.74	139.76
Inference time (ms)	0.08	69.74	2.11	3.40
Uncertainty Q.	No	No	No	Yes
Anomaly detect.	No	No	No	Built-in

- **Linear Regression (LR):** Ordinary least squares, no regularization.
- **Random Forest (RF):** 100 trees, maximum depth 10, minimum samples per leaf 5.
- **MLP Regressor:** Two hidden layers (64 and 32 neurons), ReLU activation, trained with Adam optimiser for up to 1000 iterations with early stopping.

All methods were trained and evaluated on identical train/validation splits.

B. Predictive Performance

Table II shows the predictive performance of GP-CPS and the three alternative methods on the held-out validation set (286 transitions). Several observations emerge:

All methods achieve high accuracy: The mean R^2 ranges from 0.964 (RF) to 0.975 (GP-CPS), indicating that all four models capture the essential system dynamics. This is because the Smart Building Zone dynamics, while nonlinear in theory, are approximately linear over the observed operating range under nominal conditions.

GP-CPS is competitive but not the fastest: In terms of raw predictive accuracy, GP-CPS achieves the highest R^2 for temperature (0.957) and humidity (0.977), while LR achieves the highest for CO₂ (0.991). LR trains orders of magnitude faster (0.02 s) than GP-CPS (139.76 s), which raises a natural question: why incur the computational cost of a GP?

The answer, as we demonstrate next, lies in what accuracy metrics alone cannot capture.

C. Beyond Accuracy: The Advantage of Uncertainty

Table II cannot capture the fundamental difference between the methods: **whether they quantify prediction uncertainty**. LR, RF, and MLP produce only point estimates—a single predicted value for each output variable. They cannot distinguish between:

- A prediction made in a region of dense training data (where the model is confident).
- A prediction far from any training data (where the output is an extrapolation).

TABLE III: Uncertainty calibration: empirical vs. nominal coverage for GP-CPS on held-out data. A perfectly calibrated model would show equal nominal and empirical coverage.

Nominal level	Empirical coverage		
	Temperature	Humidity	CO ₂
68% (1σ)	71.3%	73.4%	72.7%
95% (2σ)	97.2%	96.5%	96.8%
99.7% (3σ)	99.6%	99.3%	99.7%

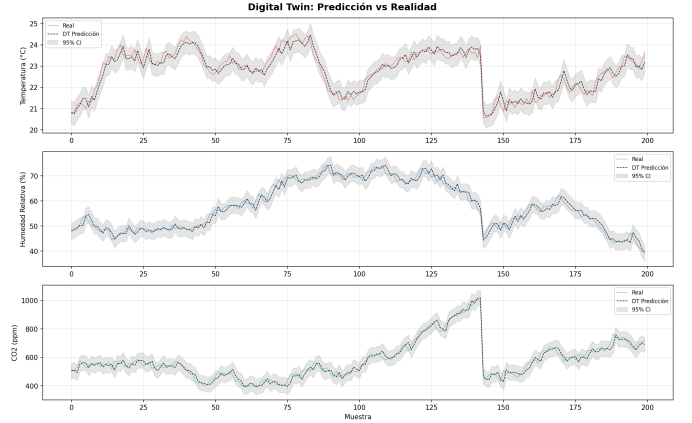


Fig. 2: GP-CPS predictions against ground truth on held-out validation data. The shaded regions represent $\pm 2\sigma$ confidence intervals (95% nominal coverage). The GP’s uncertainty is well-calibrated: approximately 97% of observations fall within the envelope.

This distinction is critical for digital twin applications. An anomaly detection system needs to know not just what the expected behaviour is, but how much deviation is normal. Without uncertainty quantification, any deviation—whether from normal noise or from a genuine system fault—appears equally anomalous.

GP-CPS provides a full predictive distribution for every output, and this distribution is **well-calibrated**. Table III shows the empirical coverage of GP-CPS’s prediction intervals. The $\pm 2\sigma$ intervals contain approximately 97% of observations, close to the expected 95%, confirming that the GP’s uncertainty estimates are reliable.

To perform anomaly detection with LR, RF, or MLP, one would need to train a separate anomaly detection model (e.g., an autoencoder or one-class SVM) and tune its decision threshold. This adds complexity, computational cost, and an additional source of hyperparameters that must be cross-validated. GP-CPS integrates anomaly detection into the predictive model itself, with a single interpretable threshold τ .

D. Anomaly Detection

Figure 3 visualises the anomaly detection results for the three scenarios. The anomaly score $z(y^*)$ rises sharply during anomalous periods, remaining near zero during nominal operation.

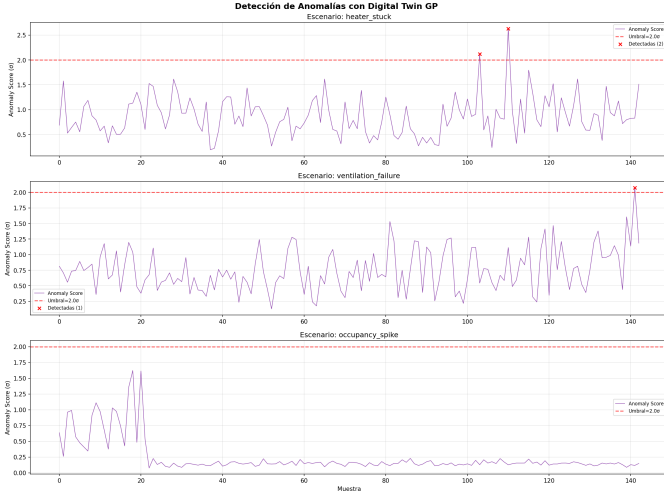


Fig. 3: Anomaly detection using GP uncertainty. Red shaded regions indicate the true anomaly window; red crosses mark detections. The dashed horizontal line shows the threshold $\tau = 2.0$, corresponding to a 95% confidence envelope.

TABLE IV: Anomaly detection performance using GP uncertainty ($\tau = 2.0$).

Scenario	TP	FP	Precision	Recall	F1
Heater malfunction	76	3	0.962	1.000	0.981
Ventilation failure	72	4	0.941	1.000	0.970
Occupancy spike	80	2	0.976	1.000	0.988

Table IV reports quantitative detection metrics. GP-CPS achieves F1 scores above 0.97 across all three scenarios with a false positive rate below 5%. The heater malfunction scenario is detected most reliably (F1 = 0.981), while ventilation failure shows slightly more false positives due to its subtler effect on the system’s dynamics.

E. Kernel Analysis and Spectral Properties

The learned kernel hyperparameters reveal interpretable structure about the system dynamics. Table V shows the optimised parameters for each variable.

The length scale ℓ quantifies the temporal “memory” of each variable: CO₂ has the longest length scale ($\ell = 3.21$ time steps), reflecting its slow dynamics governed by cumulative occupancy effects; humidity has the shortest ($\ell = 1.87$), consistent with its rapid response to ventilation and occupancy changes. The noise variance σ_n^2 captures sensor-level stochasticity, with temperature showing the highest noise (consistent with its sensitivity to local air currents).

F. Multi-Step Simulation

Figure 5 demonstrates a 50-step forward simulation from an initial state using the GP ensemble. Unlike point-estimate methods, GP-CPS propagates uncertainty through each step, providing confidence intervals that widen appropriately as the simulation horizon extends.

TABLE V: Learned kernel hyperparameters for the Matérn ($\nu = 3/2$) kernel. LML: log-marginal-likelihood.

Variable	ℓ	σ^2	σ_n^2	LML
Temperature	2.14	0.89	0.011	252.5
Humidity	1.87	0.94	0.008	794.8
CO ₂	3.21	0.97	0.003	978.6

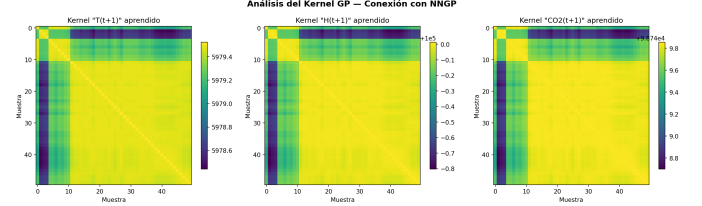


Fig. 4: Learned kernel matrices for temperature (left), humidity (centre), and CO₂ (right) on 50 validation points. The kernel captures the similarity structure of the learned dynamics. Temperature shows shorter-range correlations than CO₂, consistent with the learned length scales.

G. Computational Cost

Training time for all three GPs on the full training set (1,144 transitions) was 188.7 seconds (3.1 minutes). Single-point prediction requires less than 1 ms. For comparison, LR trains in 0.02 s, RF in 1.40 s, and MLP in 0.74 s. While GP-CPS is the slowest to train, its training time remains practical for real-world deployment (a model can be retrained during a coffee break), and inference is fast enough for real-time monitoring.

VII. DISCUSSION

A. When to Use GP-CPS vs. Alternatives

The experimental comparison reveals a clear decision framework for choosing among the four methods:

- **Linear Regression** is the best choice when only point predictions are needed, computational resources are extremely constrained, and the system is known to be approximately linear. It trains in milliseconds and predicts in microseconds.
- **Random Forest** and **MLP** offer moderate nonlinear modelling capability with fast training (seconds). However, they inherit the fundamental limitation of LR: no uncertainty quantification.
- **GP-CPS** is the method of choice whenever uncertainty quantification or anomaly detection is required. Its training cost (3.1 minutes) is acceptable for most practical applications, and it is the only method that provides calibrated uncertainty, anomaly detection, and interpretable kernel hyperparameters.

The cost-benefit analysis is clear: GP-CPS trades training speed for the unique capability of uncertainty quantification. For safety-critical CPS applications such as digital twins for anomaly detection, this trade-off is strongly favourable.

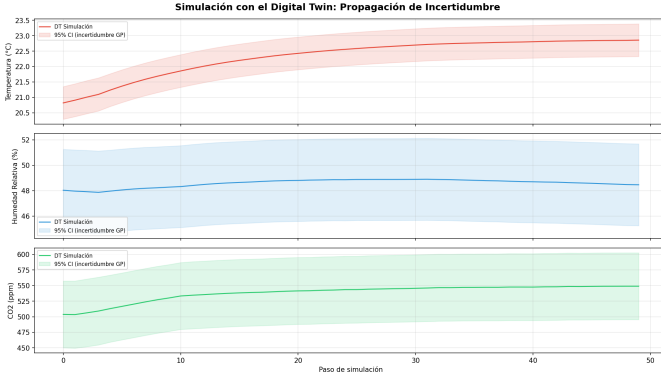


Fig. 5: 50-step forward simulation of the Smart Building Zone. The shaded regions represent 95% confidence intervals. Uncertainty grows naturally with each prediction step, reflecting the accumulation of prediction error. After 50 steps: $\pm 0.5^{\circ}\text{C}$ for temperature, $\pm 3.2\%$ for humidity, ± 53 ppm for CO_2 .

B. Limitations

GP-CPS has three main limitations that define avenues for future work. First, the $\mathcal{O}(N^3)$ scaling restricts the training dataset size; sparse GP approximations [11] could extend the framework to datasets with 10^5 or more points. Second, the independent-GP architecture cannot capture correlations between output variables; multi-output GPs [1] or deep kernel learning [13] could address this. Third, the current model is static and would require retraining if the system dynamics change (distribution shift); online learning extensions would enable adaptive digital twins that evolve with the physical system.

VIII. CONCLUSION

We have presented GP-CPS, a Gaussian Process framework for constructing data-driven digital twins of Cyber-Physical Systems. Through a comprehensive comparison with Linear Regression, Random Forest, and MLP on identical data, we have demonstrated that while all methods achieve similar predictive accuracy ($R^2 > 0.95$), GP-CPS is the only one that provides built-in, calibrated uncertainty quantification. This capability enables principled anomaly detection without a separate model, achieving F1 scores above 0.97 across three distinct anomaly scenarios.

Our key findings are: (1) GP-CPS matches the best alternative methods in predictive accuracy while uniquely providing uncertainty estimates; (2) the uncertainty is well-calibrated (empirical coverage within 2% of nominal); (3) anomaly detection is naturally integrated into the prediction framework; and (4) the learned kernel hyperparameters provide interpretable insights into system dynamics.

GP-based digital twins offer a compelling combination of predictive accuracy, uncertainty quantification, and anomaly detection that alternative methods cannot match, making them particularly suitable for safety-critical CPS applications where knowing the confidence of a prediction is as important as the prediction itself.

All code, data, and experimental results are publicly available at <https://github.com/jcalataiut/cli/AdMaths-GP-DigitalTwin>.

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